Data Analytics Term Project

Non-English Speaking Rate by State: Where Does Texas Rank?

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Data Analytics Term Project Report

1. Project Overview

This report documents our Data Analytics term project, prepared according to the course requirements for Define a Project, Implementation, and Presentation. It covers the dataset, our research question and analytic approach, the R implementation, the results with interpretation, and references.

Component	Details
Course	Data Analytics (R)
Group	Group 10 — Seohyun Choi, Zitong Zhu
Tools	R, dplyr, tidyr, ggplot2
Submission items	R script (term_project_analysis.R), this report (Word), presentation slides (PPT)

2. Define a Project

2.1 What is our data set about?

The dataset comes from the US Census Bureau (data.census.gov), table B16001 — "Language Spoken at Home by Ability to Speak English for the Population 5 Years and Over," from the 2024 American Community Survey (ACS) 5-Year Estimates.

It covers 52 regions (the 50 U.S. states, the District of Columbia, and Puerto Rico). For each region, it reports the estimated number of people (age 5+) who speak a given language at home, cross-tabulated with their self-reported English-speaking ability ("speak English very well" vs. "speak English less than very well"), along with a 90% margin of error (MOE) for every estimate.

2.2 Data Cleaning and Size

The raw CSV, as downloaded from data.census.gov, has 128 rows (one "Total" row, one "Speak only English" row, and 21 language groups each split into a language-total row and two English-ability sub-rows) and 105 columns (1 label column + 52 regions × 2 columns each for Estimate and Margin of Error).

Our research question only requires two rows per region — the overall population Total and the Speak only English count — since:

Non-English speaking population = Total − Speak only English

Cleaning steps performed in R:

1. Strip the byte-order-mark (BOM) and rename the first column to label.
2. Remove the leading non-breaking-space indentation Census uses to encode row hierarchy, using sub("^[^A-Za-z]*", "", label).

3. Filter down to the Total: and Speak only English rows only.
4. Drop all Margin of Error columns (not needed for this descriptive analysis) and keep Estimate columns only.
5. Reshape from wide (one column per state) to long, then back to wide with one row per state, using tidyr::pivot_longer() and pivot_wider().

After cleaning, the working dataset is 52 rows × 5 columns (state, total, english_only, non_english, pct_non_english) — far below the 2,000-observation ceiling, and easily processed on a laptop.

2.3 Research Question and Learning Model

Research question: "Among the 52 regions (50 states + DC + Puerto Rico), what is Texas's rank in the percentage of the population that speaks a language other than English at home?"

Learning model: This is primarily a Data Wrangling / Data Manipulation task combined with Descriptive Statistics — we compute a derived rate per region, rank all 52 regions, and summarize the distribution (mean, median, standard deviation, min, max, and Texas's percentile) to characterize where Texas stands. As an applied improvement (Section 3.10), we additionally ran a supporting one-sample t-test to confirm the ranking difference is statistically significant, rather than relying on the descriptive summary alone.

2.4 Expectations

Before running the analysis, we expected Texas to rank near the top of the 52 regions. Texas shares a long border with Mexico, has one of the largest Hispanic/Latino populations of any U.S. state, and is home to major immigrant gateway cities (Houston, Dallas, San Antonio, El Paso). Based on general demographic knowledge, we expected Texas to land in the top 5, likely behind Puerto Rico (an overwhelmingly Spanish-speaking territory) and California (the largest and one of the most diverse state populations).

2.5 Evaluation Method

Because this is a descriptive/data-wrangling project rather than a predictive model, "correctness" was evaluated as follows:

- Internal consistency check: for every state, total = english_only + non_english was verified by construction (the code derives non_english by subtraction), and all pct_non_english values fall within the valid 0–100% range.
- Sanity check against known demographics: states with well-documented large immigrant/Hispanic populations (California, Texas, New Jersey, New York, Florida) were expected to appear near the top, and states with smaller foreign-born populations (West Virginia, Montana, Mississippi) were expected to appear near the bottom — which matched the computed ranking.
- Reproducibility: the full R script runs end-to-end without errors from raw CSV to final ranked table and charts, confirming the pipeline logic is correct.

3. Implementation

3.1 Tools and Libraries

Implemented entirely in R using the tidyverse-style packages taught in class: dplyr (data manipulation: filter, mutate, group_by-style aggregation, arrange), tidyr (pivot_longer / pivot_wider for reshaping), and ggplot2 (visualization), connected with the %>% pipe.

3.2 Step 1 — Load and Clean the Data

```
lang.dat <- read.csv("./data/language_by_state.csv",
                     header = TRUE, check.names = FALSE,
                     stringsAsFactors = FALSE)
lang.dat <- as_tibble(lang.dat)
names(lang.dat)[1] <- "label"

lang.dat <- lang.dat %>%
  mutate(label = sub("^[^A-Za-z]*", "", label),
         label = trimws(label))

lang.clean <- lang.dat %>%
  filter(label %in% c("Total:", "Speak only English"))
```

3.3 Step 2 — Reshape Wide to Long, then to a Tidy Table

```
lang.long <- lang.clean %>%
  select(label, ends_with("!!Estimate")) %>%
  pivot_longer(cols = -label, names_to = "state", values_to = "value") %>%
  mutate(state = gsub("!!Estimate", "", state),
         value = as.numeric(gsub(",", "", value)))

lang.wide <- lang.long %>%
  pivot_wider(names_from = label, values_from = value) %>%
  rename(total = `Total:`, english_only = `Speak only English`)
```

3.4 Step 3 — Calculate the Non-English Speaking Rate

```
lang.ratio <- lang.wide %>%
  mutate(non_english = total - english_only,
         pct_non_english = round(non_english / total * 100, 2)) %>%
  select(state, total, english_only, non_english, pct_non_english)
```

3.5 Step 4 — Rank All 52 Regions and Locate Texas

```
lang.rank <- lang.ratio %>%
  arrange(desc(pct_non_english)) %>%
  mutate(rank = row_number()) %>%
  select(rank, state, pct_non_english, total, non_english)

texas.rank <- lang.rank %>% filter(state == "Texas")
```

3.6 Step 5 — Visualization (Texas Highlighted)

Two ggplot2 bar charts were produced: (1) all 52 regions ranked from highest to lowest non-English speaking rate, and (2) a zoomed-in view of the top 15 regions with percentage labels, for use in the presentation slides. In both charts Texas is highlighted in a distinct color against the other states.

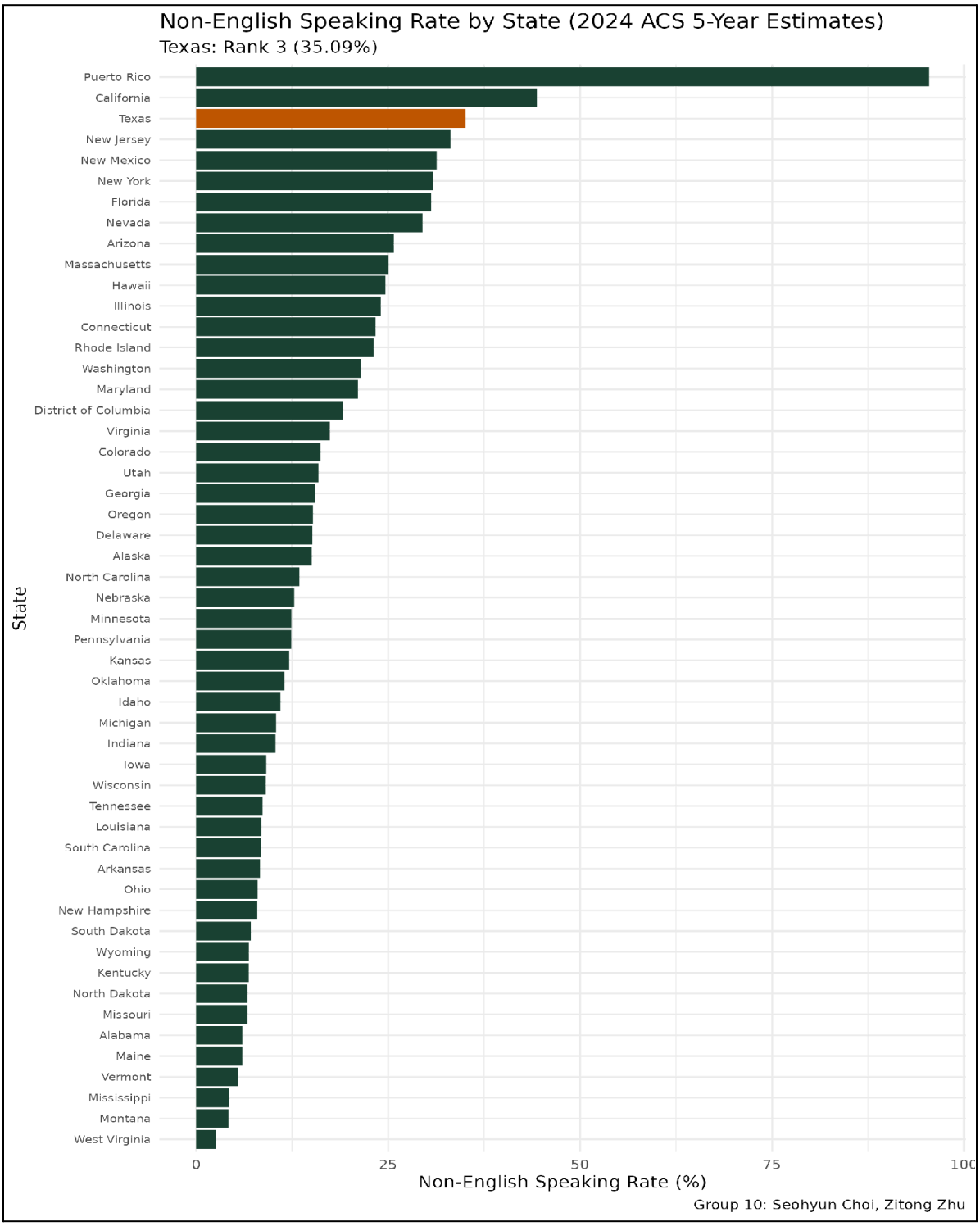


Figure 1. Non-English speaking rate, all 52 regions (ranked, Texas highlighted).

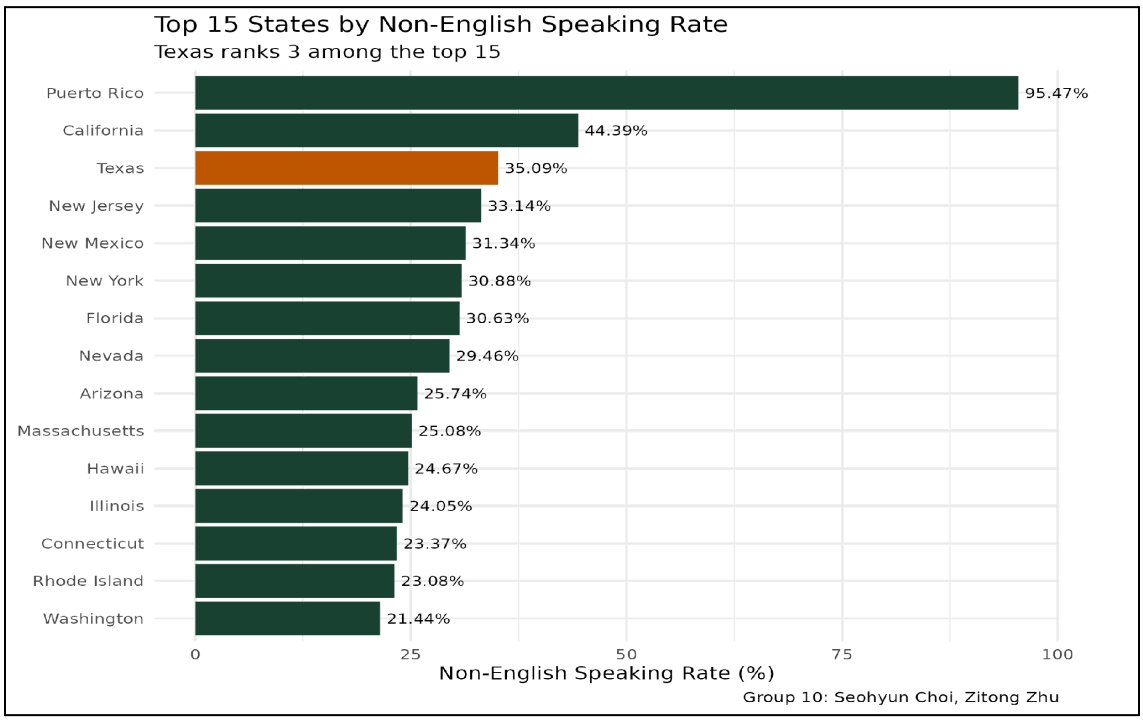


Figure 2. Top 15 regions by non-English speaking rate, with percentage labels.

3.7 Additional Visualizations: Distribution, Region, and Population

Beyond the ranked bar charts, we added three more chart types covered in the course's ggplot2 material (histogram, boxplot, and scatter plot) to look at the result from different angles.

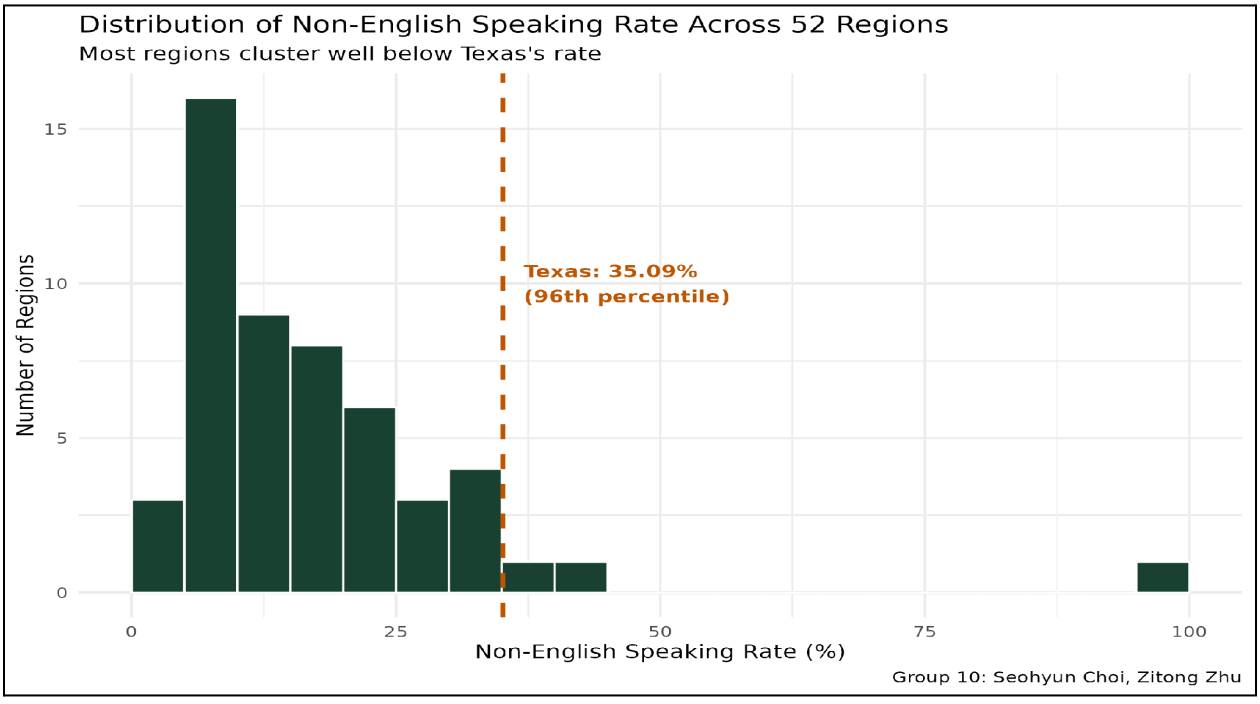


Figure 3. Histogram of the non-English speaking rate across all 52 regions, with Texas marked.

The histogram (geom_histogram + geom_vline) shows that most regions cluster between 5% and 25%, with a long right tail. Texas sits well out in that tail, and Puerto Rico is the extreme outlier near 95%. This gives a visual complement to the 96th-percentile statistic reported in Section 3.8.

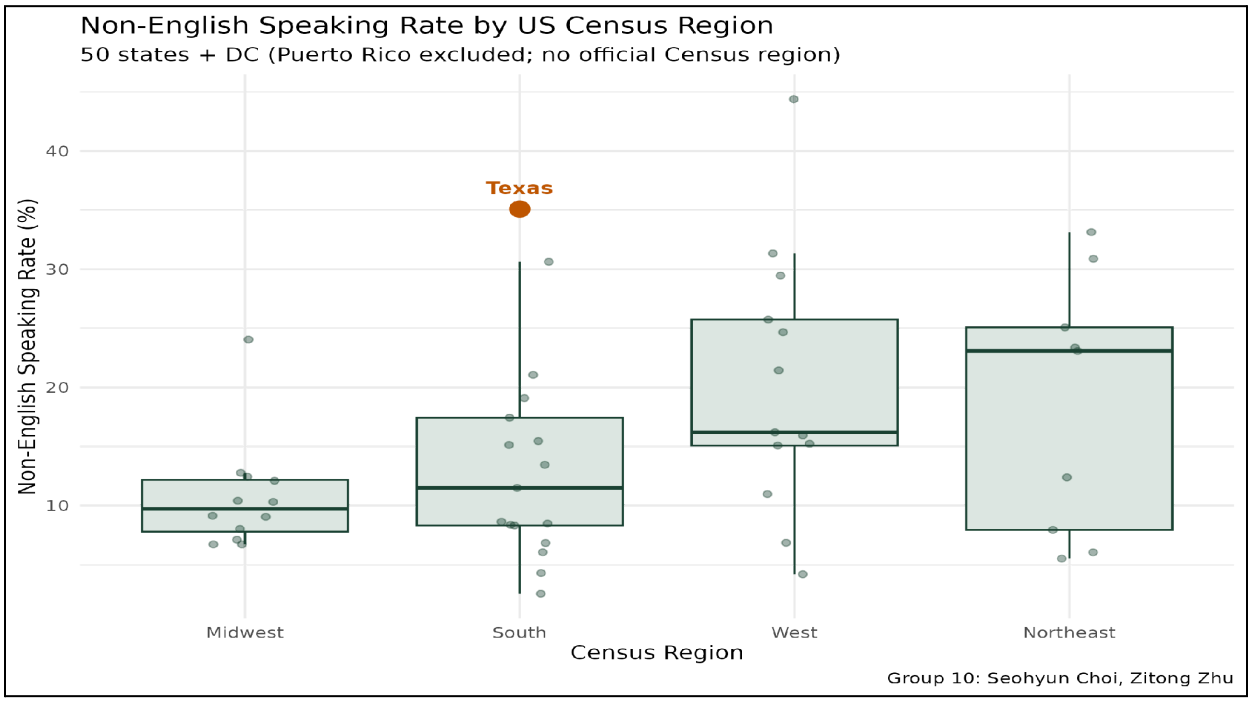


Figure 4. Non-English speaking rate by official US Census Bureau region (boxplot), Texas marked.

We assigned each of the 50 states + DC to its official Census Bureau region (Northeast, Midwest, South, West); Puerto Rico has no official region and was excluded from this comparison. The boxplot (geom_boxplot) shows the South region has the widest spread, and Texas is a clear outlier even within its own region — only California (in the West) reaches a comparably high rate.

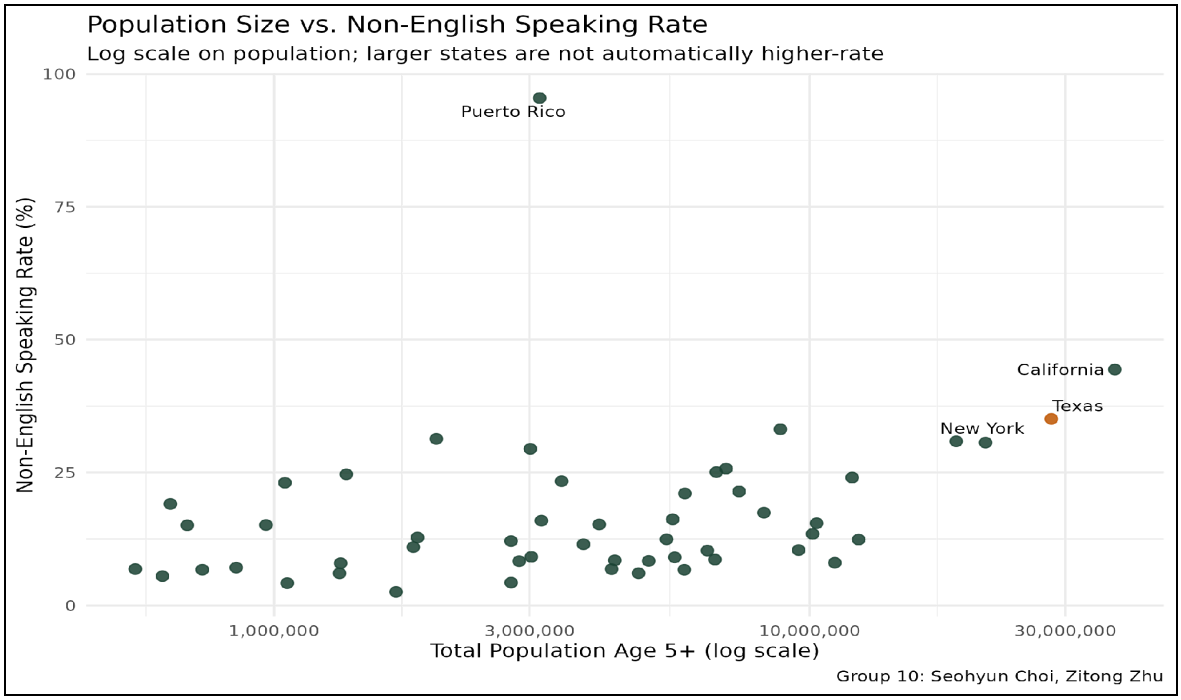


Figure 5. Total population vs. non-English speaking rate (log scale on population), Texas marked.

The scatter plot (geom_point, log-scaled x-axis) checks whether the non-English speaking rate is simply a byproduct of state size. It is not a strong relationship — many small and mid-sized states sit anywhere from near 0% to over 30%, and while Texas, California, and New York (three of the largest states) do sit toward the higher end, some very large states are not automatically high-rate. Population size alone does not explain Texas's ranking.

3.8 Results

Texas ranks #3 out of 52 regions, with 35.09% of its population age 5+ speaking a language other than English at home (9,913,837 out of 28,253,216 people).

Rank	Region	Non-English Speaking Rate
1	Puerto Rico	95.47%
2	California	44.39%
3	Texas	35.09%
4	New Jersey	33.14%
5	New Mexico	31.34%
6	New York	30.88%
7	Florida	30.63%

Descriptive statistics across all 52 regions:

Statistic	Value
Mean	17.0%
Median	12.6%
Standard deviation	14.6
Minimum	2.54% (West Virginia)
Maximum	95.47% (Puerto Rico)
Texas's percentile	96.2th percentile

3.9 Interpretation

- Texas ranks in the top 3 of 52 regions and sits at the 96.2 percentile — only Puerto Rico and California have a higher non-English speaking rate.
- Texas's rate (35.09%) is roughly double the 52-region average (17.0%) and nearly triple the median (12.6%), confirming our pre-analysis expectation.
- The top-ranked regions (Puerto Rico, California, Texas, New Jersey, New Mexico, New York, Florida) share large Hispanic/Latino populations and/or high levels of immigration — several are U.S.–Mexico border states or major immigrant-gateway states.
- The lowest-ranked regions (West Virginia, Montana, Mississippi, etc.) are inland/rural states with comparatively low immigrant population shares and less population mobility.

- This supports our initial hypothesis: Texas's border with Mexico and large Spanish-speaking population plausibly explain its high national ranking.
- Figure 4 adds a nuance: Texas isn't just high nationally, it's an outlier even compared to other Southern states — most of the South clusters between about 8% and 17%, while Texas sits at 35%.
- Figure 5 shows that population size alone doesn't explain the pattern — the relationship between state size and non-English speaking rate is weak, so Texas's high rate reflects its specific demographic makeup (border geography, Hispanic population share), not simply the fact that it's a large state.

3.10 Applied Improvement: One-Sample t-test

The assignment specifically asks us to apply an improvement idea, not just list one, so we implemented the first item from our improvement list: a formal hypothesis test checking whether Texas's rate is statistically distinguishable from the typical region, rather than relying on the ranking alone.

We treated the other 51 regions' rates as a sample and tested whether their true mean is below Texas's specific value (35.09%):

- H_0 (null): the mean non-English speaking rate of the other 51 regions equals Texas's rate (35.09%).
- H_1 (alternative): the mean of the other 51 regions is less than Texas's rate (one-sided).
- Significance level: $\alpha = 0.05$.

```
other.pct <- lang.rank %>%
  filter(state != "Texas") %>%
  pull(pct_non_english)

texas.ttest <- t.test(other.pct,
                      mu = texas.rank$pct_non_english,
                      alternative = "less")
```

Statistic	Value
Mean of other 51 regions	16.68%
Texas's rate (test value, mu)	35.09%
t-statistic	−9.057
Degrees of freedom	50
p-value	2.02×10^{-12}
95% CI for other regions' mean	(−∞, 20.08%)

Because $p = 2.02 \times 10^{-12}$ is far below $\alpha = 0.05$, we reject H_0. The 95% confidence interval places the true mean of the other 51 regions at or below 20.08% — still well under Texas's 35.09%.

This gives us a statistical, not just descriptive, basis for our headline claim: Texas's non-English speaking rate is significantly higher than the typical U.S. region, not simply the result of natural variation across 52 regions.

3.11 Possible Further Improvements

Given more time, the analysis could be extended in the following ways:

- Break the “non-English” category down by specific language (Spanish, Vietnamese, Chinese, etc.) to see which languages drive Texas's ranking specifically, rather than treating all non-English speakers as one group.
- Incorporate the Margin of Error columns (currently dropped) to build confidence intervals around each state's rate and test whether Texas's rank versus its close neighbors (New Jersey, New Mexico) is statistically distinguishable.
- Pull the data directly via the tidycensus R package / Census API instead of a manually exported CSV, which would make the pipeline reusable for other ACS tables or years.
- Add a choropleth (state-shaded) map using the sf or usmap package for a more geographic view of the pattern shown in Figures 1–2.

3.12 References

- U.S. Census Bureau. (2024). Table B16001: Language Spoken at Home by Ability to Speak English for the Population 5 Years and Over. 2024 American Community Survey 5-Year Estimates. Retrieved from data.census.gov.
- Wickham, H., et al. dplyr: A Grammar of Data Manipulation. https://dplyr.tidyverse.org
- Wickham, H., et al. tidyr: Tidy Messy Data. https://tidyr.tidyverse.org
- Wickham, H. ggplot2: Elegant Graphics for Data Analysis. https://ggplot2.tidyverse.org
- Course materials: Data Analytics course lecture slides and dplyr/ggplot2 examples (Lecture01–Lecture10, DataManipulationWithR.pdf, DataVisualiztionWithRggplot.pdf), Instructor: Kia Teymourian.

4. Presentation Summary

This report, together with the accompanying slide deck, serves as the presentation of our project. In the slides we walk through: the dataset and why we chose it, the research question, the cleaning/wrangling steps, the final ranking with Texas highlighted, the descriptive statistics, and our interpretation — written to be understandable to any reader in the data analytics field, not only R specialists.

Key takeaway for the presentation: Texas ranks #3 out of 52 U.S. Census regions in non-English speaking rate (35.09%), behind only Puerto Rico and California, and well above the 52-region average of 17.0%.